LOW-POWER CAN-FD CONTROLLER WITH ENHANCED FAULT RECOVERY FOR AUTOMOTIVE SOCS

AGENDA

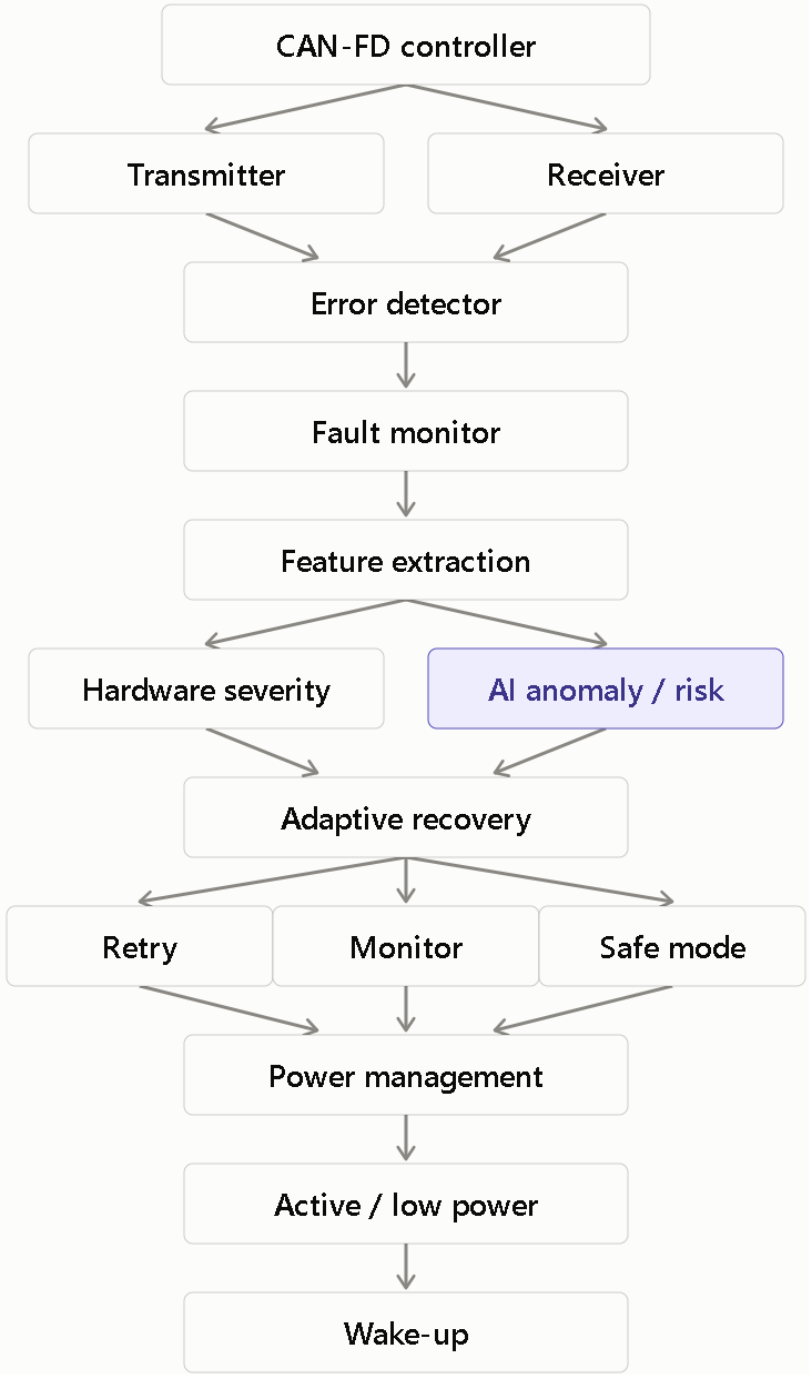
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- INTRODUCTION
- LITERATURE SURVEY AND BACKGROUND RESEARCH
- PROBLEM IDENTIFICATION
- OBJECTIVES
- EXISTING SYSTEM
- PROPOSED SYSTEM
- SYSTEM ARCHITECTURE / BLOCK DIAGRAM
- METHODOLOGY AND TECHNICAL APPROACH
- RTL DESIGN AND IMPLEMENTATION
- AI-BASED ANOMALY DETECTION
- FUNCTIONAL VERIFICATION AND SIMULATION
- SYNTHESIS AND RTL-TO-GDSII IMPLEMENTATION
- EXPECTED RESULTS
- CONTRIBUTION TO SPECIFIC SUSTAINABLE DEVELOPMENT GOALS
- REFERENCES

INTRODUCTION

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- Modern vehicles contain multiple Electronic Control Units (ECUs) that require reliable and high-speed communication.
- CAN-FD (Controller Area Network – Flexible Data-Rate) extends Classical CAN by supporting larger payloads and higher data-phase communication rates.
- This project focuses on the design of a **configurable CAN-FD controller using System Verilog** with enhanced fault recovery and adaptive low-power operation.
- The controller includes **error monitoring, fault recovery, configurable bit timing, flexible payload handling, and low-power control**.
- An **AI-based anomaly detection module** is also proposed to identify abnormal CAN-FD traffic using communication features such as error counts, retransmissions, message frequency, CAN ID, and DLC.
- The design will be verified through RTL simulation and targeted for **RTL-to-GDSII implementation** with timing, area, and power analysis.



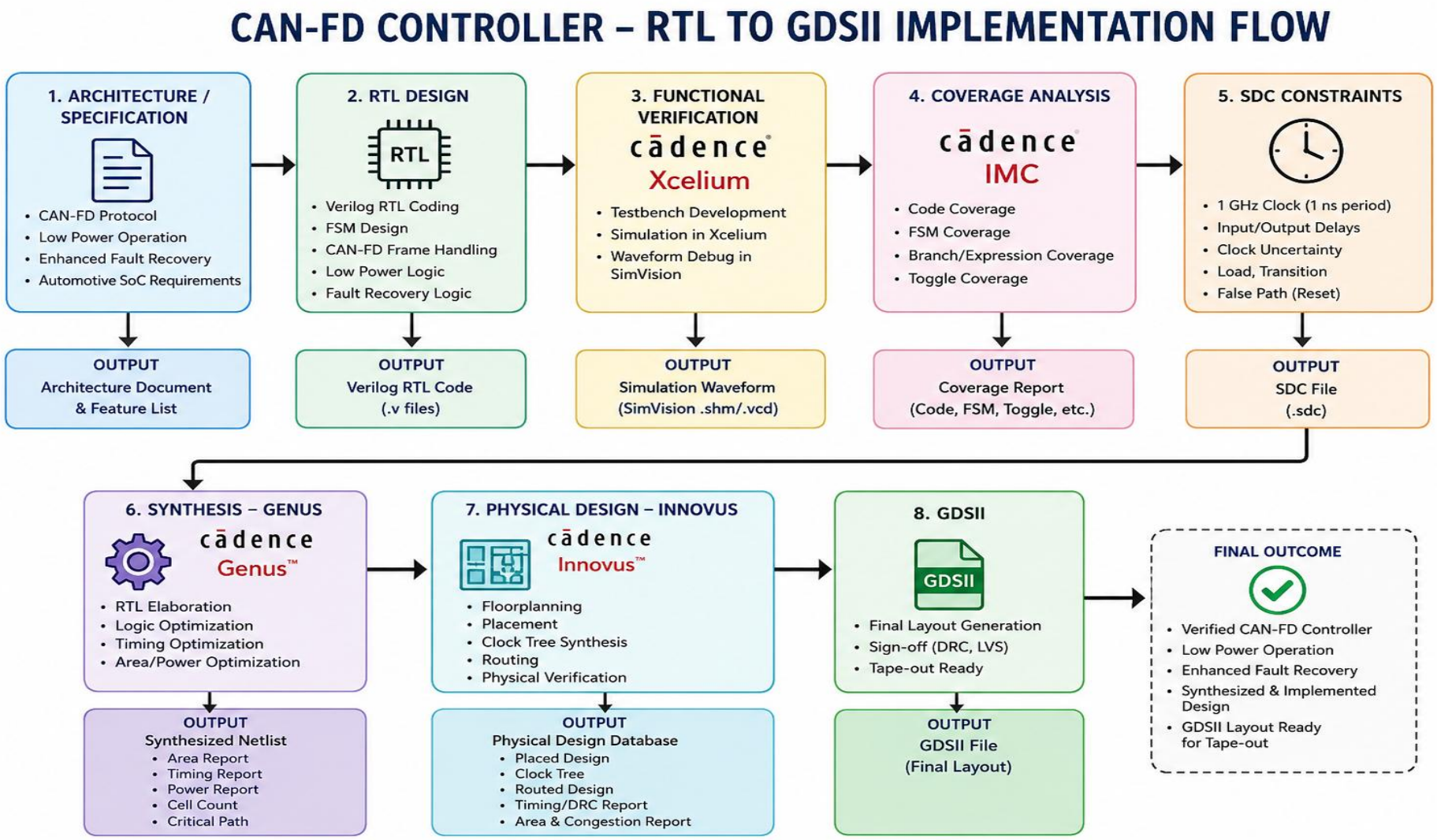
1. **Error Detection:** Finds errors in CAN-FD communication and monitors faults.

2. **Feature Extraction:** Collects important information to identify hardware problems and AI-based risks.

3. **Adaptive Recovery:** Takes suitable action, such as retrying, monitoring, or switching to safe mode.

4. **Power Management:** Controls power usage by switching between active mode, low-power mode, and wake-up mode.

SYNTHESIS AND RTL-TO-GDSII IMPLEMENTATION



LITERATURE SURVEY

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

TITLE	PUBLICATION DETAILS	WHAT WE UNDERSTOOD / LEARNT	MERITS	DEMERITS
Configurable CAN FD Controller: Design, Implementation and Evaluation	**Authors:** M.E.G.A. Sin **Publication:** 2022, Doctoral Dissertation, Middle East Technical University	• Configurable CAN-FD Controller Architecture & Parameter Selection • CAN-FD Frame Handling: Arbitration, Control, Data & CRC • Controller Verification through Functional Testing • Need for Power Optimization in CAN-FD Controllers • Scope for Enhanced Error Detection & Fault Recovery	• Provides a configurable architecture suitable for different CAN-FD requirements. • Directly useful as a reference for designing a CAN-FD hardware controller.	• Limited emphasis on power optimization, as low-power operation is not a primary design objective of the proposed controller. • Limited focus on advanced fault-recovery mechanisms, leaving scope for improving the controller's resilience under communication fault conditions.

LITERATURE SURVEY

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

TITLE	PUBLICATION DETAILS	WHAT WE UNDERSTOOD / LEARNT	MERITS	DEMERITS
Development of a CAN FD Communication-Based HOD Pad Function Inspection System	**Authors:** Wu Sung Choi, Jae-Wan Park, Jun-Hee Lee, Myung-Hoon Sung, Jae-Won Kim, Young-Gyu Jeong, Beom-Seok Kwon, Hyeon-Bin Bae, Jae Woo, Jeong-Hoon Jeong, Teak-Young Seung. **Publication:** *Journal of Electrical Engineering & Technology*, 2026. DOI: 10.1007/s42835-026-02839-0.	• Configurable TX/RX Mailbox Architecture • Acceptance Filtering for RX Messages • SPI-Based Host MCU Interface • Efficient Embedded Memory Utilization • CAN-FD Controller FPGA Implementation & Evaluation	• Provides systematic functional inspection and monitoring of HOD pad systems. • Improves fault identification and supports safety verification in automotive applications.	• Focuses on an automotive inspection system, rather than designing a configurable CAN-FD hardware controller. • Does not specifically address low-power CAN-FD operation or enhanced fault-recovery architecture, leaving scope for hardware-level improvements.

LITERATURE SURVEY

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

TITLE	PUBLICATION DETAILS	WHAT WE UNDERSTOOD / LEARNT	MERITS	DEMERITS
Explainable & Deterministic Intrusion Detection for CAN-FD: A Logic Extraction Framework	**Authors:** G. Rithvika and R. Radha **Publication:** *Scientific Reports*, 2026	• CAN-FD Traffic Parsing & Feature Extraction • CART-Based Logic Extraction for Intrusion Detection • Deterministic Boolean Rules for Attack Detection • Volumetric, Protocol & Semantic Attack Detection • Low-Latency Logic for Resource-Constrained Automotive Systems	• Improves the security and interpretability of CAN-FD intrusion detection. • Deterministic logic can support consistent and explainable detection decisions.	• Focuses primarily on intrusion detection and security, rather than CAN-FD controller hardware design. • Does not specifically address low-power operation or enhanced fault recovery in the controller architecture.

LITERATURE SURVEY

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

TITLE	PUBLICATION DETAILS	WHAT WE UNDERSTOOD / LEARNT	MERITS	DEMERITS
Controller Area Network With Flexible Data Rate (CAN FD) Eye Diagram Prediction	**Authors:** Junyong Park, Man-Ho Lee, Shinyoung Park, Jonghoon Kim, Joungho Kim, DongHyun Kim **Publication:** 2024 Journal: *IEEE Transactions on Electromagnetic Compatibility* Volume: 66, Issue 3, pp. 949–959	• The paper studies the signal behavior of CAN-FD communication at different data rates. • It develops a method to predict the CAN-FD eye diagram and compares the prediction with measured results. • We understood that high-speed CAN-FD operation requires reliable signal integrity, which is important for robust automotive communication.	•Helps analyze CAN-FD communication at higher data rates. • Considers signal integrity and physical communication behavior. • Useful for understanding reliable high-speed CAN-FD operation.	• Mainly focuses on physical-layer signal behavior. • Does not design a configurable CAN-FD controller. • Does not address RTL-level fault recovery or low-power controller architecture.

LITERATURE SURVEY

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

TITLE	PUBLICATION DETAILS	WHAT WE UNDERSTOOD / LEARNT	MERITS	DEMERITS
Analysis of Cooperative Operating Concepts for Robust Data Transmission in Periodically Disturbed CAN FD Networks	**Authors:** Carina Austermann, Stephan Frei **Publication:** 2024 Journal: *IEEE Transactions on Electromagnetic Compatibility*	• The paper investigates robust CAN-FD communication under electromagnetic disturbances. • Automotive CAN-FD networks can experience communication problems due to EMI and coupling in vehicle wiring. • We understood that a CAN-FD controller should be designed to detect communication problems and maintain reliable operation under fault conditions.	• Focuses on reliability of CAN-FD communication. • Considers realistic automotive electromagnetic disturbances. • Supports the need for robust communication mechanisms.	• Mainly studies communication robustness at the network/physical level. • Does not implement an RTL CAN-FD controller. • Does not focus on dynamic power optimization.

PROBLEM IDENTIFICATION

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- Conventional CAN controllers have limited support for flexible CAN-FD communication.
- CAN-FD requires efficient handling of payloads up to **64 bytes**.
- Different communication requirements require configurable bit timing and frame parameters.
- Continuous hardware activity can result in unnecessary dynamic power consumption.
- Bit, CRC, stuff and form errors can affect communication reliability.
- Repeated communication faults require effective fault recovery and retransmission.
- Traditional rule-based monitoring may not identify abnormal traffic patterns effectively.
- An integrated architecture combining **CAN-FD control, fault recovery, low-power management and AI-based anomaly detection** is required.

OBJECTIVE

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

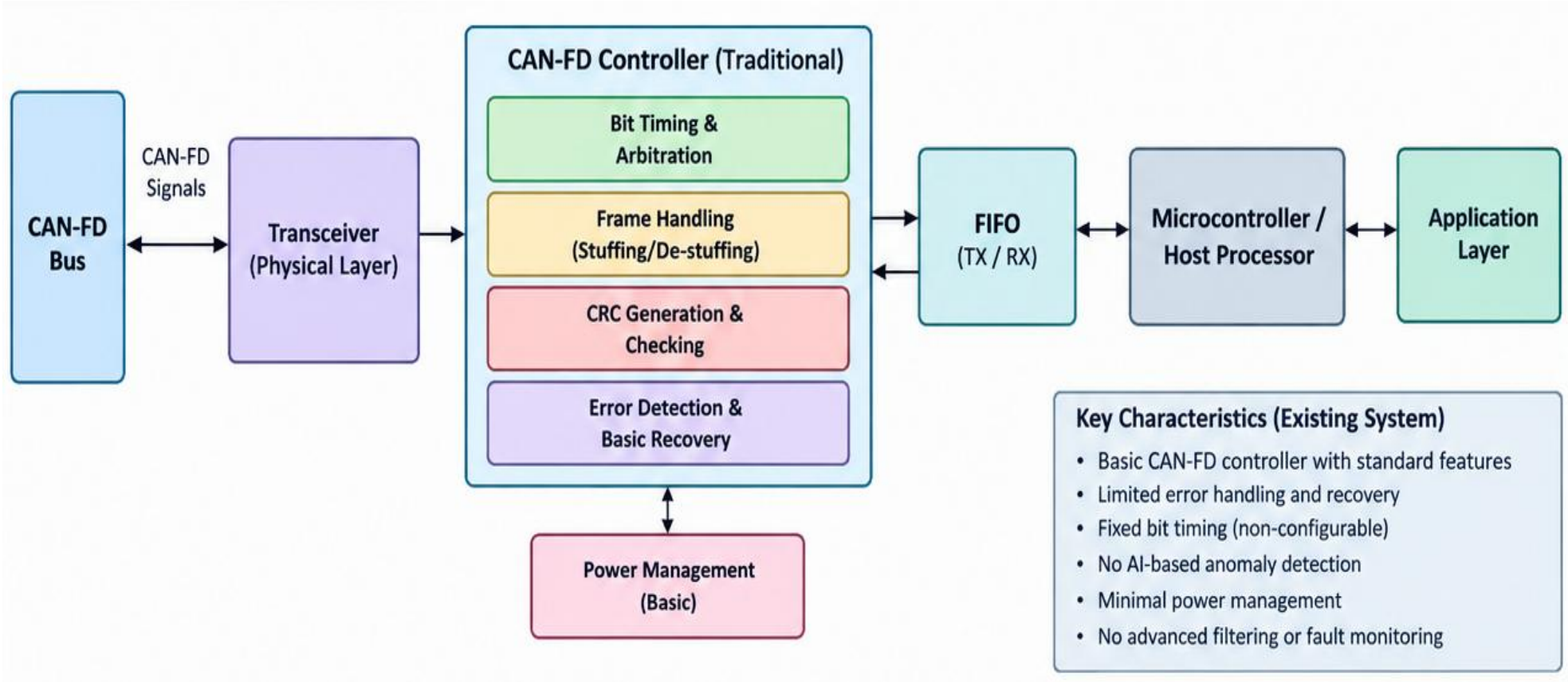
- To design a **configurable CAN-FD controller using System Verilog**.
- To support flexible CAN-FD payload handling up to **64 bytes**.
- To implement configurable nominal and data-phase bit timing.
- To detect communication errors including CRC, ACK, bit, stuff and form errors.
- To develop an enhanced fault recovery mechanism with retransmission and safe recovery.
- To reduce unnecessary switching activity using adaptive low-power control.
- To extract CAN-FD communication features for anomaly detection.
- To train a lightweight **Decision Tree model using Python and Scikit-learn**.
- To implement the trained decision rules as a **System Verilog AI inference block**.
- To verify the complete design and perform RTL-to-GDSII analysis.

EXISTING SYSTEM

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- Classical CAN controllers primarily support conventional CAN communication.
- Payload size is smaller compared with CAN-FD.
- CAN-FD data-phase advantages are not available in conventional CAN.
- Error detection is available, but recovery capability depends on the controller architecture.
- Limited configurability for different communication requirements.
- Continuous hardware activity can increase dynamic switching power.
- Conventional controllers generally do not include AI-based traffic anomaly detection.
- Fault monitoring, recovery and power management may exist as separate functions rather than a unified architecture.

EXISTING SYSTEM – Block Diagram / Flow Chart



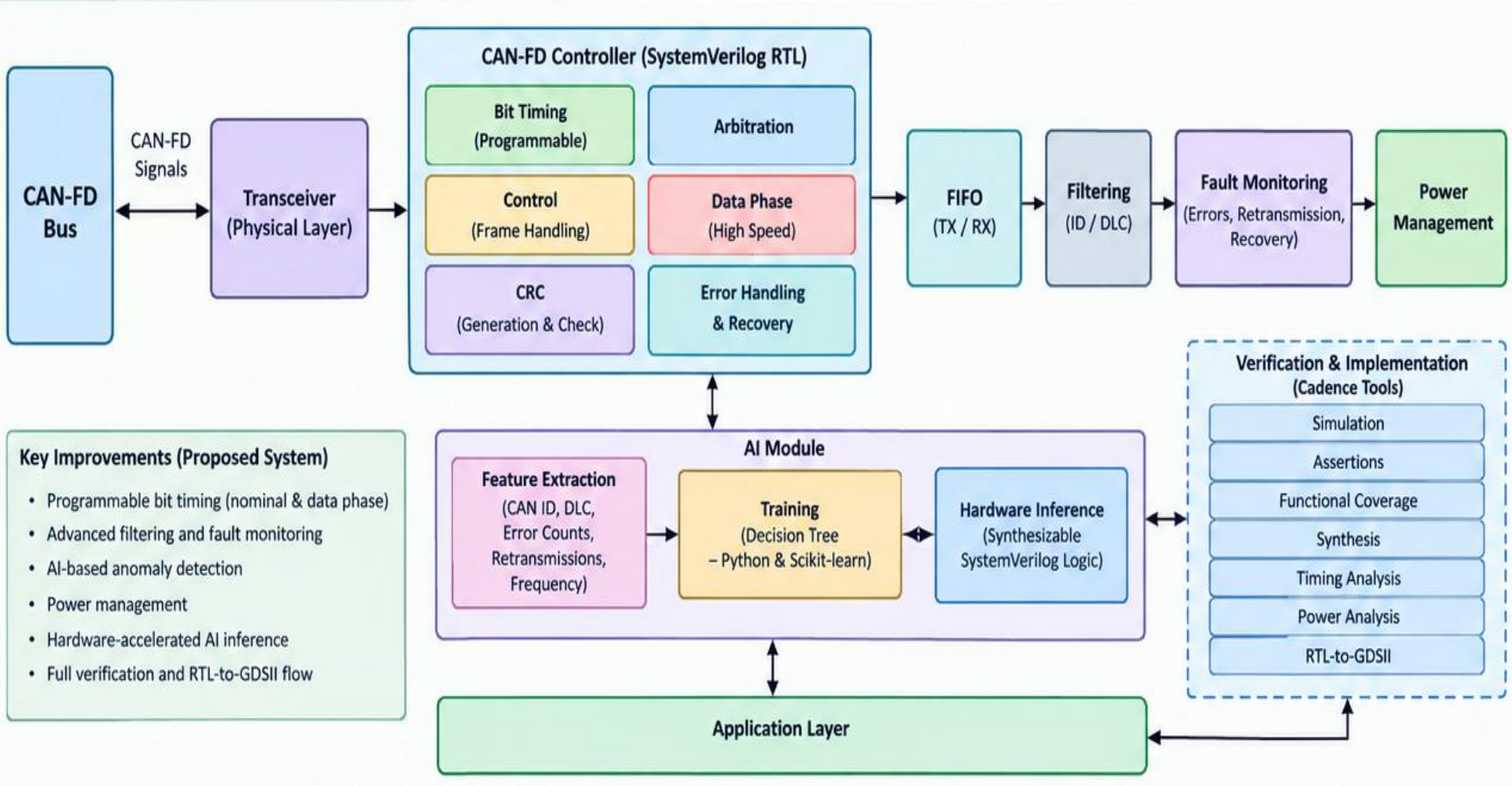
PROPOSED SYSTEM

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- CAN-FD frame transmission and reception
- Configurable bit timing
- Payload handling up to 64 bytes
- Arbitration and frame control
- CRC and communication error detection
- TX/RX FIFO management
- CAN ID acceptance filtering
- Enhanced fault recovery and retransmission
- Adaptive Active/Idle/Sleep/Wake operation
- CAN-FD traffic feature extraction
- Decision Tree based anomaly detection
- Risk-based recovery control
- RTL-to-GDSII implementation

PROPOSED SYSTEM – Block Diagram / Flow Chart

KiTE
KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius



METHODOLOGY AND TECHNICAL APPROACH

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

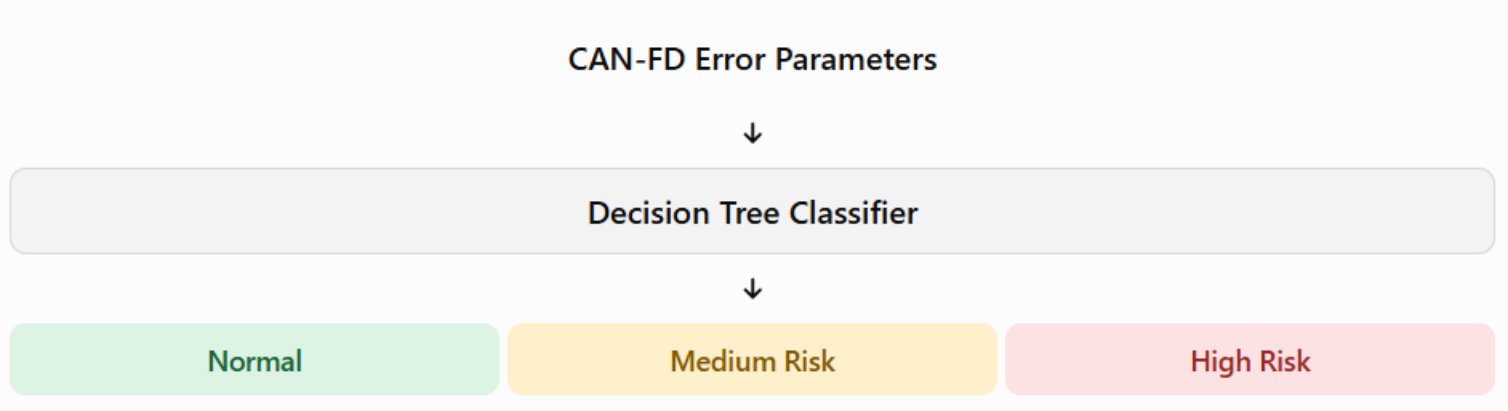
- **CAN-FD Specification**
  - Define frame structure and communication parameters
- **RTL Design**
  - CAN-FD TX/RX with error monitoring and fault recovery
- **Feature Extraction**
  - CAN ID, DLC, errors, retransmission and frequency
- **AI Detection**
  - Decision Tree training in Python
  - Implement trained rules in System Verilog
- **Verification**
  - Simulation, assertions, coverage and waveform analysis
- **RTL-to-GDSII**
  - Synthesis → Physical Design → Timing → DFT → GDSII

AI-Based Fault Prediction in CAN-FD Controller

Purpose: To predict the fault risk in the CAN-FD communication system.

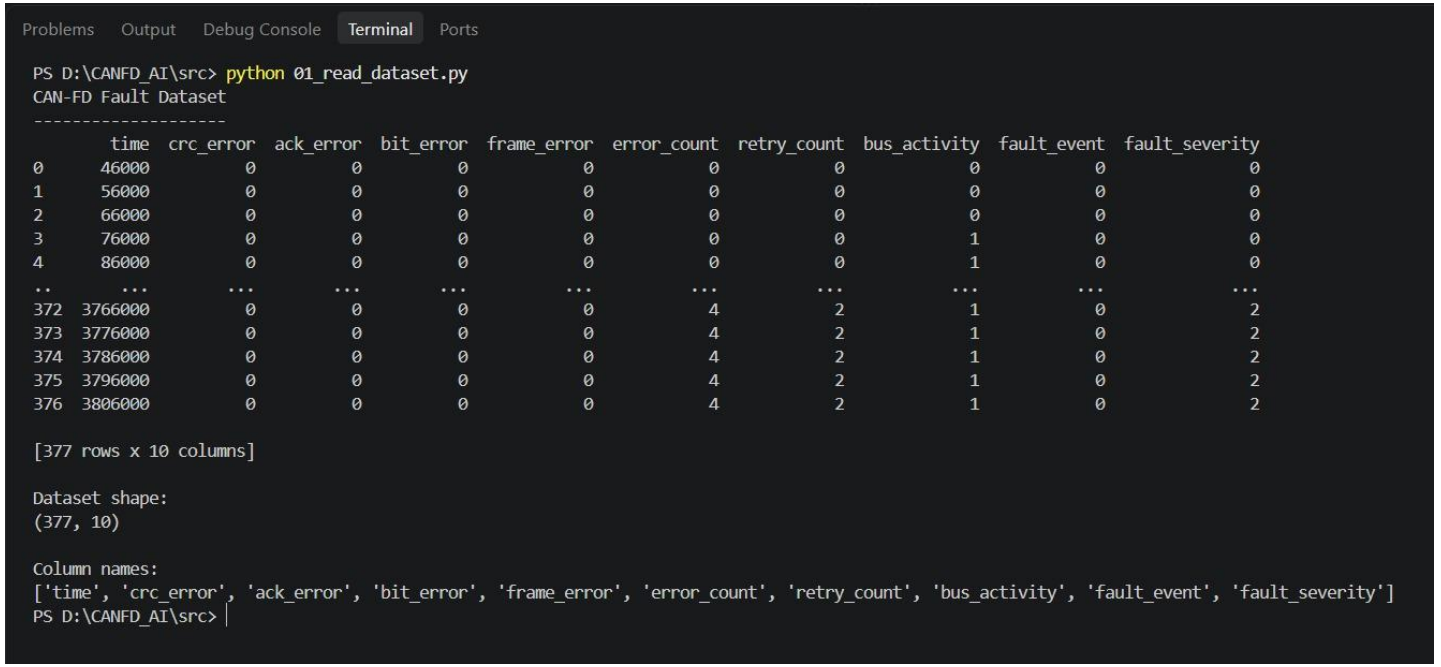
Algorithm: Decision Tree classifier is used for fault risk classification.

Input Parameters: CRC error, ACK error, bit error, frame error, error count, retry count, and bus activity.



Fault Recovery: The predicted risk helps the controller select a suitable recovery action.

Implementation: Python and Machine Learning are used to develop and test the AI model.



Dataset Loading and Reading (01_read_dataset.py)

- The Python program loads the CAN-FD fault dataset for AI model development.
- It reads the dataset and displays the number of rows and columns.
- It identifies the input features and target labels used for fault risk classification.
- The dataset is used as the input for training and testing the Decision Tree classifier.

Purpose: To verify that the dataset is loaded correctly before developing the AI model.

```
PS D:\CANFD_AI\src> python 02_train_model.py
Dataset loaded successfully.
Number of samples: 377

Training samples: 282
Testing samples: 95

Model training completed.

Model Accuracy:
1.0

Model saved successfully.
```

AI Model Training and Testing (02_train_model.py)

- Dataset: 377 samples used for AI model development.
- Data Splitting: 282 samples for training and 95 samples for testing.
- Model Training: The Decision Tree classifier is trained using the CAN-FD fault dataset.
- Model Accuracy: 100% accuracy on the test set (95 samples).
- Model Saving: The trained model is saved for future fault risk prediction.

Purpose: To train and evaluate the AI model for classifying CAN-FD communication faults into different risk levels.

- **AI-Based Fault Risk Prediction (03_predict.py)**
- Prediction Results: The trained Decision Tree model predicts the risk level for each sample in the CAN-FD dataset.
- Predicted Risk: The first 20 samples are classified as 0, indicating a Normal condition.
- Risk Level 0 – Normal: 159 samples are classified as normal, indicating low or no fault risk.
- Risk Level 1 – Medium Risk: 101 samples are classified as medium risk, indicating a moderate possibility of communication faults.
- Risk Level 2 – High Risk: 117 samples are classified as high risk, indicating a higher level of communication fault risk.
- Total Predictions: The model classifies 377 samples into three risk categories (159 + 101 + 117 = 377).
- Final Result: The AI model identifies different risk levels in CAN-FD communication and can support suitable fault-handling and recovery decisions.
- Purpose: To predict CAN-FD fault risk levels using AI and support improved communication reliability in automotive systems.

```
PS D:\CANFD_AI\src> python 03_predict.py

Prediction Results:
    predicted_risk
0                0
1                0
2                0
3                0
4                0
5                0
6                0
7                0
8                0
9                0
10               0
11               0
12               0
13               0
14               0
15               0
16               0
17               0
18               0
19               0

Risk Level Counts:
predicted_risk
0    159
1    101
2    117
Name: count, dtype: int64
PS D:\CANFD_AI\src>
```

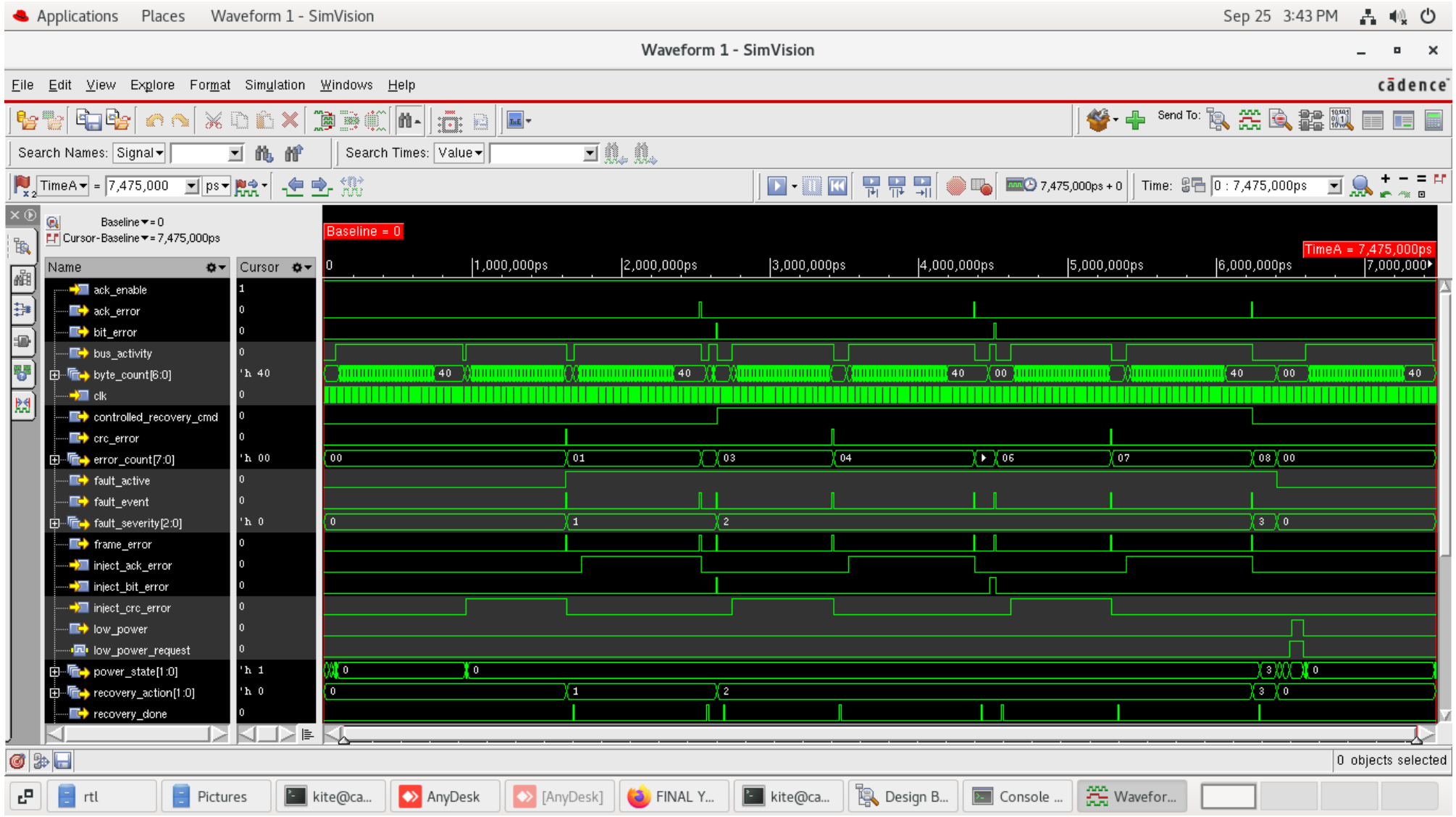
RTL DESIGN AND IMPLEMENTATION

Module	Function
CAN-FD TX/RX	Frame transmission and reception
Error Monitoring	CRC, ACK and communication-error detection
Fault Recovery	Recovery from detected faults
Power Management	Active, idle and sleep control
Feature Extraction	Generates communication features for AI
AI Inference	Software implementation of trained decision rules

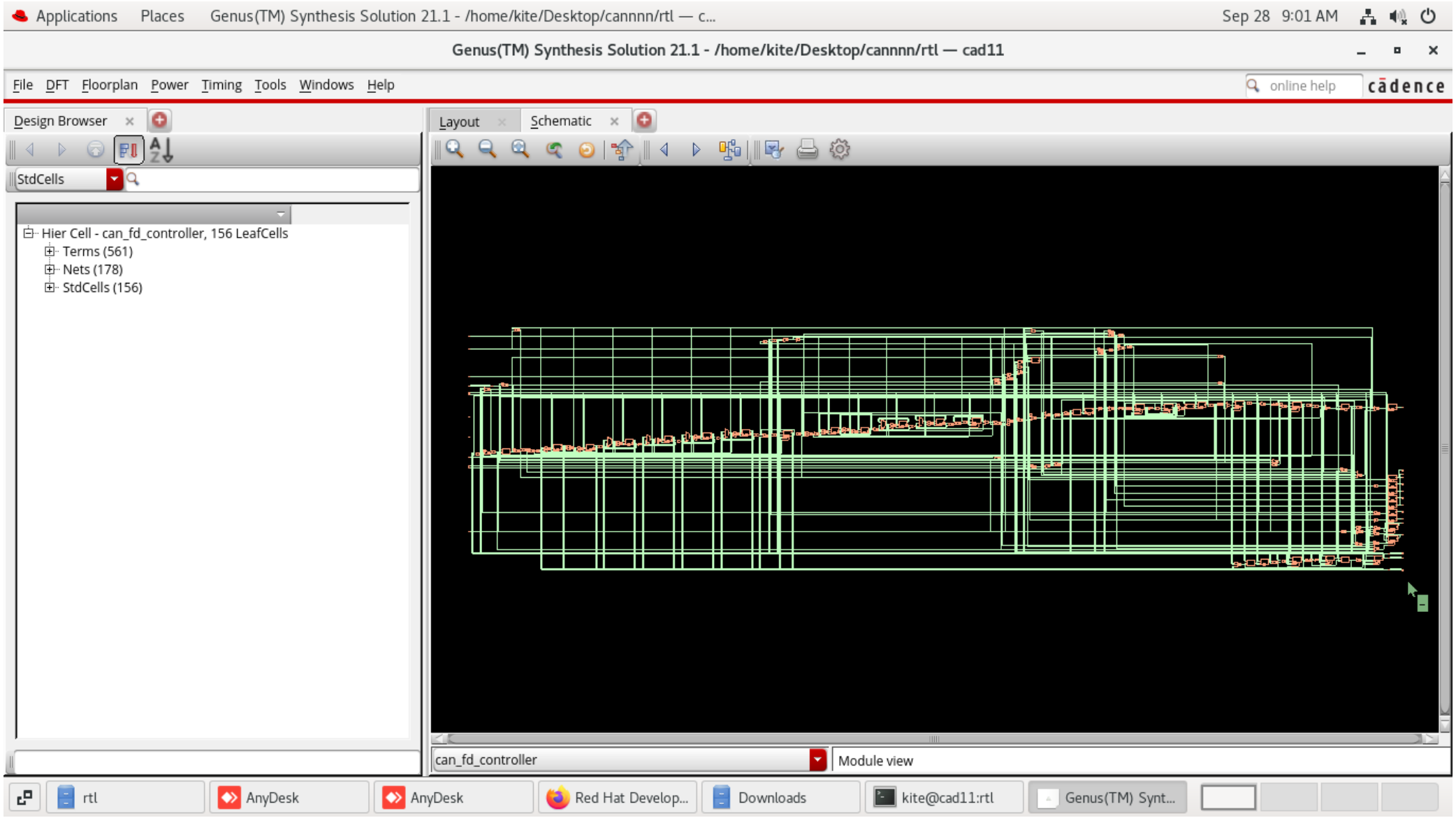
Design Specifications

- **HDL:** System Verilog
- **Payload:** Up to 64 bytes
- **Target:** Automotive SoC applications
- **Architecture:** Modular RTL design
- **Flow:** RTL → Verification → Synthesis → Physical Design → GDSII

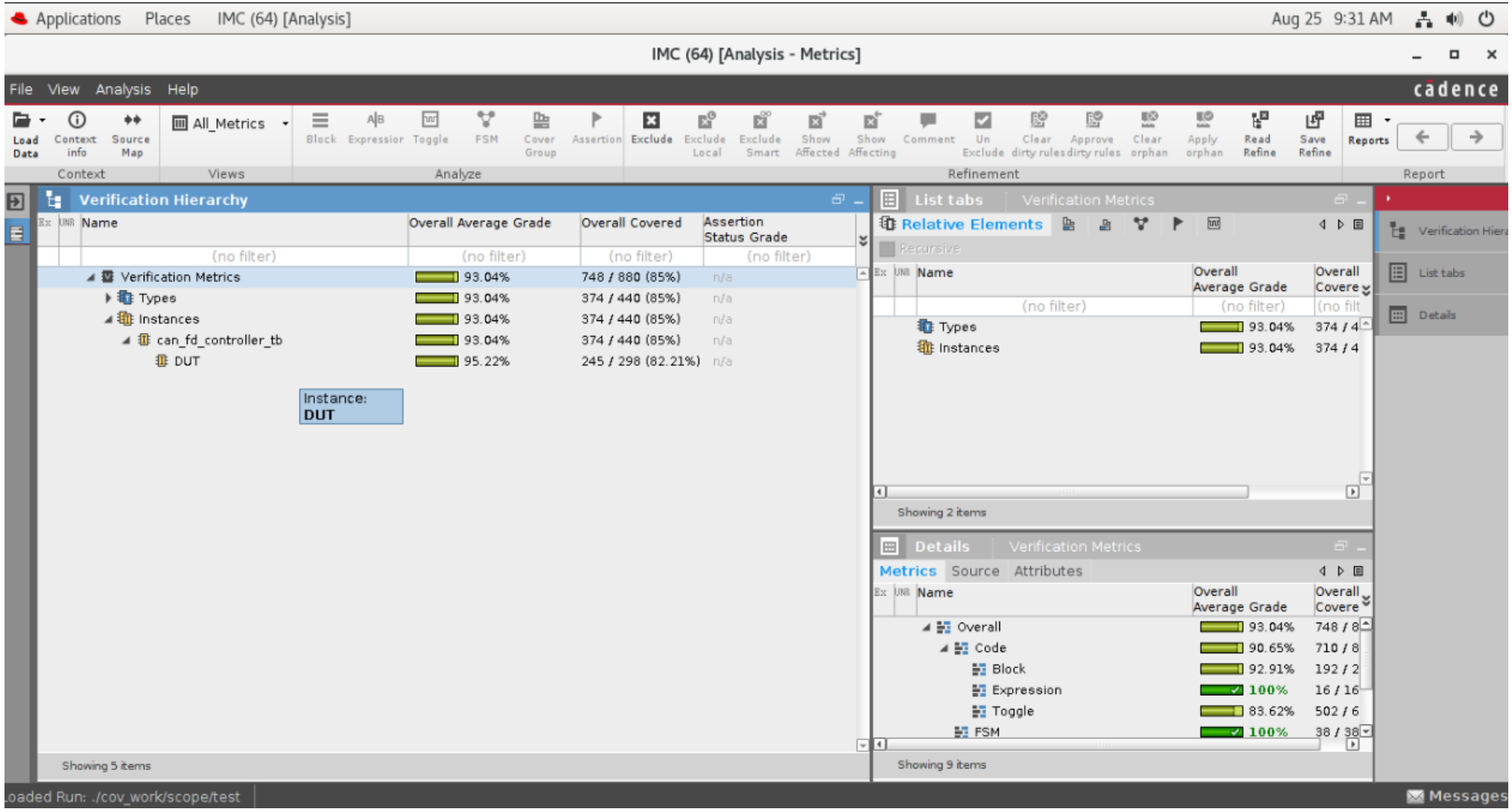
WAVEFORM - Cadence Xcelium

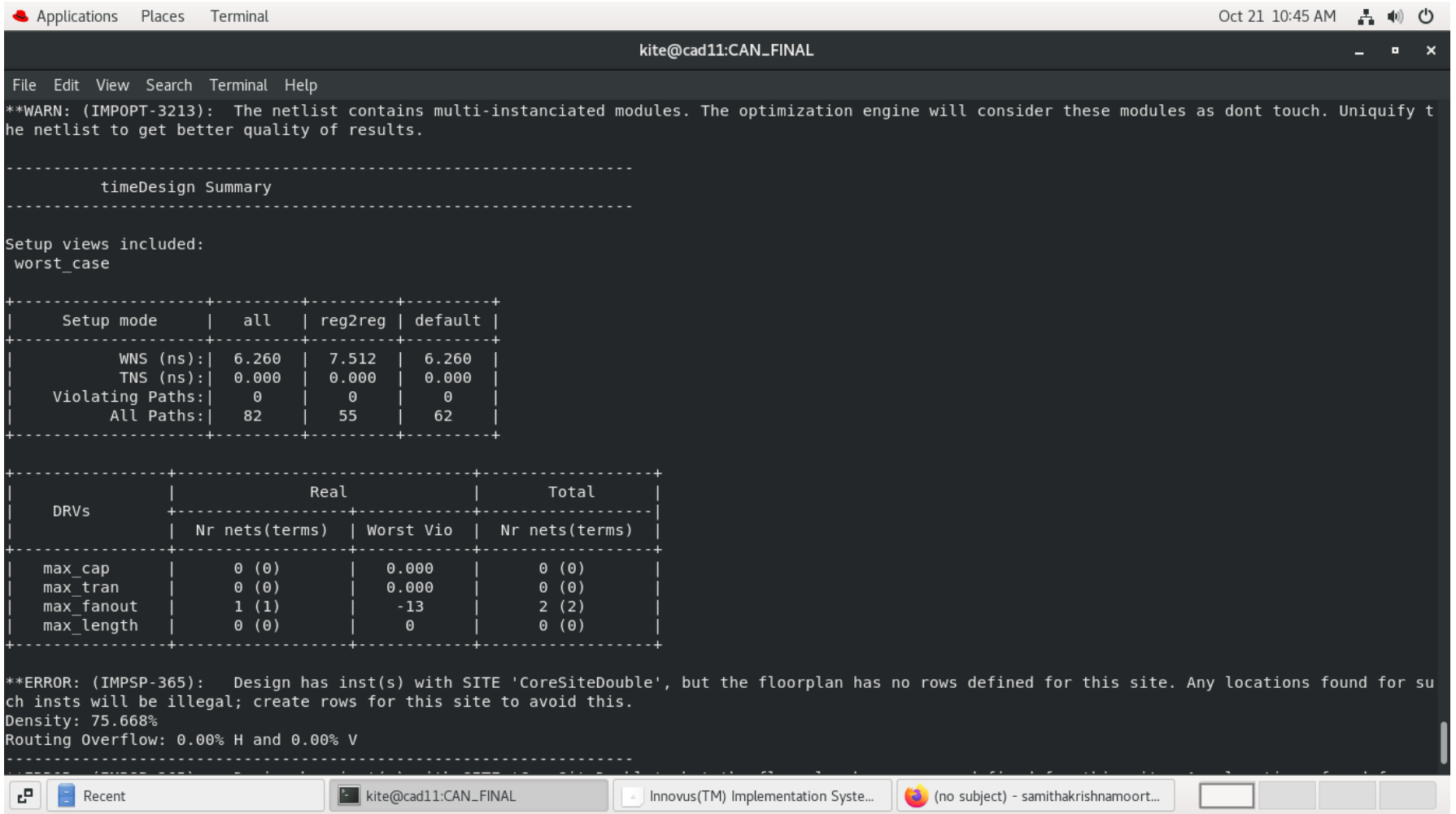


GENUS

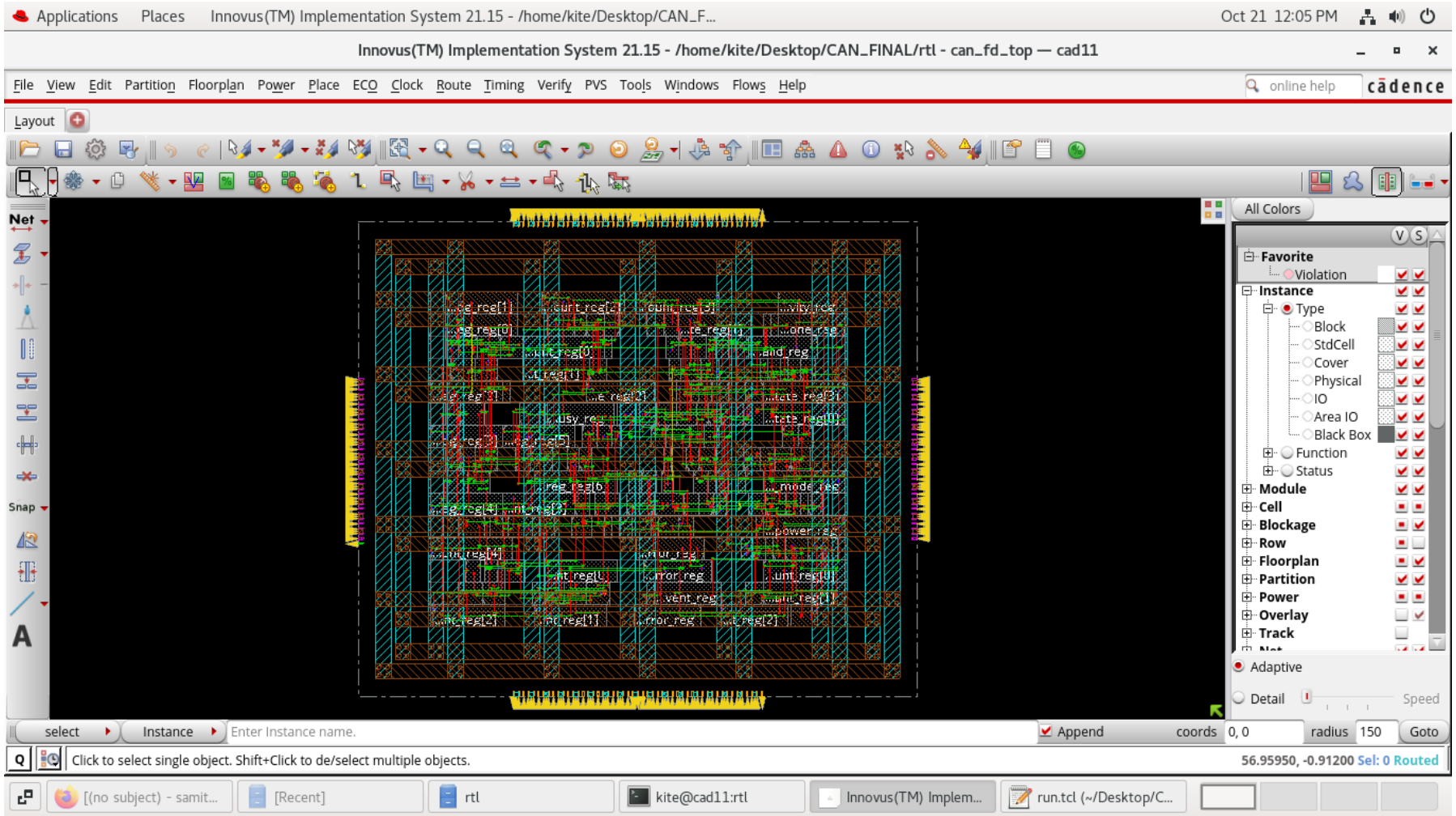

Applications
Places
Genus(TM) Synthesis Solution 21.1 - /home/kite/Desktop/cannnn/rtl — c...
Sep 28 9:01 AM
Genus(TM) Synthesis Solution 21.1 - /home/kite/Desktop/cannnn/rtl — cad11
File DFT Floorplan Power Timing Tools Windows Help
online help
cadence
Design Browser
Layout
Schematic
StdCells
Hier Cell - can_fd_controller, 156 LeafCells
Terms (561)
Nets (178)
StdCells (156)
can_fd_controller
Module view
rtl
AnyDesk
AnyDesk
Red Hat Develop...
Downloads
kite@cad11:rtl
Genus(TM) Synt...

COVERAGE


Applications
Places
IMC (64) [Analysis]
Aug 25 9:31 AM
IMC (64) [Analysis - Metrics]
File
View
Analysis
Help
cadence
Load Data
Context info
Source Map
All_Metrics
Block
Expressior
Toggle
FSM
Cover Group
Assertion
Exclude
Exclude Local
Exclude Smart
Show Affected
Show Affecting
Comment
Un Exclude
Clear dirty rules
Approve dirty rules
Clear orphan
Apply orphan
Read Refine
Save Refine
Reports
Context
Views
Analyze
Refinement
Report
Verification Hierarchy
Name
Overall Average Grade
Overall Covered
Assertion Status Grade
(no filter)
Verification Metrics
93.04%
748 / 880 (85%)
n/a
Types
93.04%
374 / 440 (85%)
Instances
93.04%
374 / 440 (85%)
can_fd_controller_tb
93.04%
374 / 440 (85%)
DUT
95.22%
245 / 298 (82.21%)
Instance:
DUT
Showing 5 items
List tabs
Verification Metrics
Relative Elements
Recursive
Overall Covere
Showing 2 items
Details
Metrics
Source
Attributes
Overall
90.65%
710 / 8
Block
92.91%
192 / 2
Expression
100%
16 / 16
Toggle
83.62%
502 / 6
FSM
38 / 38
Showing 9 items
Verification Hiera
List tabs
Details
oaded Run: ./cov_work/scope/test
Messages


Applications Places Terminal
Oct 21 10:45 AM
kite@cad11:CAN_FINAL
File Edit View Search Terminal Help
**WARN: (IMPOPT-3213): The netlist contains multi-instanciated modules. The optimization engine will consider these modules as dont touch. Uniquify the netlist to get better quality of results.
timeDesign Summary
Setup views included:
worst_case
Setup mode	all	reg2reg	default
WNS (ns):	6.260	7.512	6.260
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	82	55	62
DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	1 (1)	-13	2 (2)
max_length	0 (0)	0	0 (0)
**ERROR: (IMPSP-365): Design has inst(s) with SITE 'CoreSiteDouble', but the floorplan has no rows defined for this site. Any locations found for such insts will be illegal; create rows for this site to avoid this.
Density: 75.668%
Routing Overflow: 0.00% H and 0.00% V
Recent
kite@cad11:CAN_FINAL
Innovus(TM) Implementation Syste...
(no subject) - samithakrishnamoort...

INNOVUS


Applications
Places
Innovus(TM) Implementation System 21.15 - /home/kite/Desktop/CAN_F...
Oct 21 12:05 PM
Innovus(TM) Implementation System 21.15 - /home/kite/Desktop/CAN_FINAL/rtl - can_fd_top — cad11
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
online help
cadence
Layout
Net
Snap
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
56.95950, -0.91200 Sel: 0 Routed
[(no subject) - samit...
[Recent]
rtl
kite@cad11:rtl
Innovus(TM) Implem...
run.tcl (~/Desktop/C...

EXPECTED RESULTS

- Successful CAN-FD frame transmission and reception
- Reliable detection of communication errors
- Effective controller-level fault recovery
- AI-based identification of abnormal CAN-FD communication patterns
- Risk classification for detected anomalies
- Reduced unnecessary switching during idle conditions
- Successful System Verilog functional verification
- Improved verification through assertions and functional coverage
- Successful RTL synthesis and implementation
- Generation of final GDSII design

Performance Parameters

- **Area → Timing → Power → AI Detection → Verification Coverage**

CONTRIBUTION TO SPECIFIC SUSTAINABLE DEVELOPMENT GOALS

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

SDG 7 – Affordable and Clean Energy

- Apply low-power design techniques to minimize unnecessary dynamic switching and reduce power consumption in the CAN-FD controller.
- Enhance energy efficiency through activity-based power management and optimized RTL design.

SDG 11 – Sustainable Cities and Communities

- Support reliable data communication between automotive electronic control units to enhance transportation system reliability.
- Strengthen vehicle communication robustness through efficient fault detection, recovery mechanisms, and AI-based fault-risk prediction.

SDG 12 – Responsible Consumption and Production

- Promote energy-efficient and resource-conscious hardware design for automotive System-on-Chip (SoC) applications.
- Optimize RTL logic and hardware resource utilization to support efficient VLSI implementation and reduce unnecessary circuit complexity.

REFERENCE

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

- 1) "**Brief Industry Paper: Response Time Evaluation of Cross-Domain Communication in CAN-FD and TSN,"** in *Proc. IEEE*, 2024
- 2) **M. S. Dhanush and T. Ananthakrishna**, "Enhancing In-Vehicle Networks with CAN-FD: A Study of Protocol Improvements over Classical CAN," in *Proc. IEEE ICECCC*, 2024, doi: 10.1109/ICECCC61767.2024.10593831.
- **3) J. Park, M. Lee, S. Park, J. Kim, J. Kim, and D. H. Kim**, "Controller Area Network With Flexible Data Rate (CAN FD) Eye Diagram Prediction," *IEEE Transactions on Electromagnetic Compatibility*, vol. 66, no. 3, pp. 949–959, 2024, doi: 10.1109/TEMC.2024.3350054.
- 4) **S. S. B. et al.**, "Securing CAN FD by Implementing AES-128, SHA256, and Message Counter Based on FPGA," in *Proc. IEEE*, 2021.
- **5) "Intelligent Connected Vehicle CAN-FD Bus Network Security Protocol,"** in *Proc. IEEE*, 2023.

REFERENCE

- 6) **“A Practical Security Architecture for In-Vehicle CAN-FD,”** *IEEE*, 2021.
- 7) **“Performance-Based Fault Detection and Fault-Tolerant Control for Nonlinear Systems With T–S Fuzzy Implementation,”** *IEEE Transactions on Cybernetics*, vol. 51, no. 2, pp. 801–814, 2021, doi: 10.1109/TCYB.2019.2951534.
- 8) **“Fault Detection for Systems With Model Uncertainty and Disturbance via Coprime Factorization and Gap Metric,”** *IEEE Transactions on Cybernetics*, vol. 52, no. 8, pp. 7765–7775, 2022, doi: 10.1109/TCYB.2021.3051368.
- 9) **“Fault Detection for Nonlinear Dynamic Systems With Consideration of Modeling Errors: A Data-Driven Approach,”** *IEEE Transactions on Cybernetics*, vol. 53, no. 7, pp. 4259–4269, 2023, doi: 10.1109/TCYB.2022.3163301.
- 10) **“Performance-Based Hierarchical Fault-Tolerant Control for Closed-Loop Systems With Multiplicative Faults: A Data-Driven Design Method,”** *IEEE Transactions on Cybernetics*, vol. 54, no. 11, pp. 6593–6606, 2024, doi: 10.1109/TCYB.2024.3397807.

KGiSL INSTITUTE OF TECHNOLOGY
co-Kreate your Genius

THANK YOU